

LANDING PAGE



LOGIN PAGE



Enter Phone Number

We'll send you a verification code

Phone Number

+91

Enter 10-digit number

Send OTP

OR

New User

Admin Login

Demo: Use phone 9999999999 for admin access



Enter Phone Number

We'll send you a verification code

Phone Number

+91

8877143049

Send OTP

OR

New User

Admin Login

Demo: Use phone 9999999999 for admin access

OTP VERIFICATION



Verify OTP

Enter the 6-digit code sent to +91 8877143049

Verification Code

Verify OTP

Change Phone Number



Verify OTP

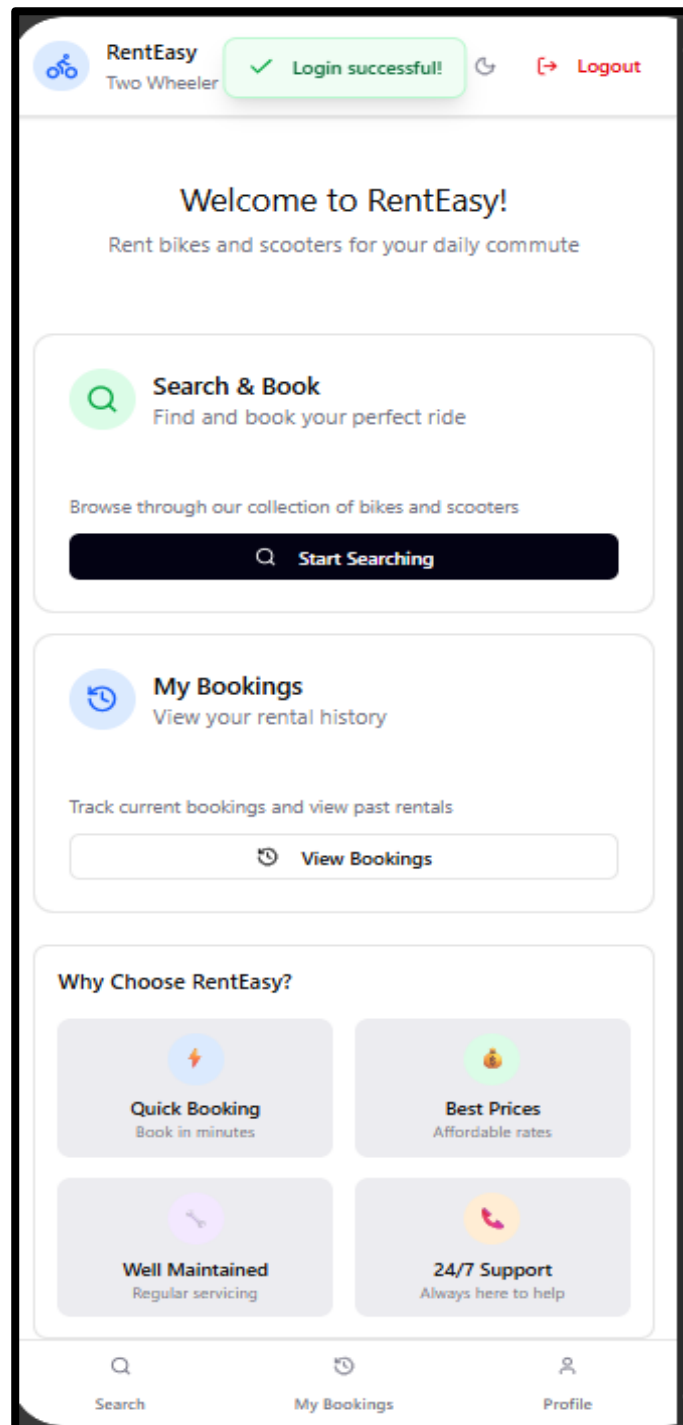
Enter the 6-digit code sent to +91 8877143049

Verification Code

Verify OTP

Change Phone Number

USER DASHBOARD



REGISTER NEW USER



Register New User

Create a new account with your phone number

Phone Number

+91

Enter 10-digit number

Send OTP for Registration

Back to Login

Demo: Use phone 9999999999 for admin access



Verify OTP

Enter the 6-digit code sent to +91 8877143049

Verification Code

Enter 6-digit OTP

Verify OTP

Change Phone Number

NEW USER REGISTRATION



Complete Registration

Fill in your details to complete registration

Full Name *

Enter your full name

Age *

Enter your age

Gender

Select gender

Driving License Number *

Enter your driving license number

Complete Registration

Start Over

* Required fields

FORM VALIDATION



Complete Registration

Fill in your details to complete registration

Full Name *

Jatin Kumar Soni

Age *

17

Gender

Male

Driving License Number *

ASDF454565

Complete Registration

Start Over

* Required fields

Age is required and must be at least 18 years

USER PROFILE

←

My Profile

Edit Profile

Please upload your driving license and Aadhar card to complete your profile. These documents are required for vehicle bookings.

Personal Information

Full Name

Jatin Kumar Soni

Phone Number

8877143049

Verified

Email

jatinkumarsoni@example.com

Age

18

Gender

Male

Address

Enter your address here

Documents

Driving License

License Number

ASDF454565

Document

Not uploaded

←

My Profile

Edit Profile

Personal Information

Full Name

Jatin Kumar Soni

Phone Number

8877143049

Verified

Email

jatinkumarsoni@example.com

Age

18

Gender

Male

Address

Enter your address here

Documents

Driving License

License Number

ASDF454565

Document

Not uploaded

Aadhar Card

Aadhar Number

Not set

Document

Not uploaded

→

Logout

EDIT PROFILE

←

My Profile

✕ Cancel

💾 Save

ⓘ

Please upload your driving license and Aadhar card to complete your profile. These documents are required for vehicle bookings.

Personal Information

👤

Full Name

Jatin Kumar Soni

📞

Phone Number

8877143049

Verified

✉

Email

jatinkumarsoni@example.com

📅

Age

18

👤

Gender

Male

▼

📍

Address

Enter your address here

Documents

📄

Driving License

License Number

←

My Profile

✕ Cancel

💾 Save

jatinkumarsoni@example.com

📅

Age

18

👤

Gender

Male

▼

📍

Address

Enter your address here

Documents

📄

Driving License

License Number

ASDF454565

Document

📶

Upload

Upload a clear photo of your driving license. Max file size: 5MB

📄

Aadhar Card

Aadhar Number

Enter 12-digit Aadhar number

Document

📶

Upload

Upload a clear photo of your Aadhar card. Max file size: 5MB

➔ Logout

ADMIN LOGIN



Enter Phone Number

We'll send you a verification code

Phone Number

+91

9999999999

Send OTP

Demo: Use phone 9999999999 for admin access



Verify OTP

Enter the 6-digit code sent to +91 9999999999

Verification Code

123456

Verify OTP

Change Phone Number

Demo: Use phone 9999999999 for admin access

DASHBOARD



Admin Panel
Vehicle Rental
Management

  Profile  Logout

Admin Dashboard

Manage vehicles and bookings



4
Total Vehicles



0
Active Bookings



Vehicle Management

Manage your fleet of bikes and scooters

Add New

Edit Details

Remove



Manage Vehicles



Booking Management

Review and manage customer booking requests

Pending

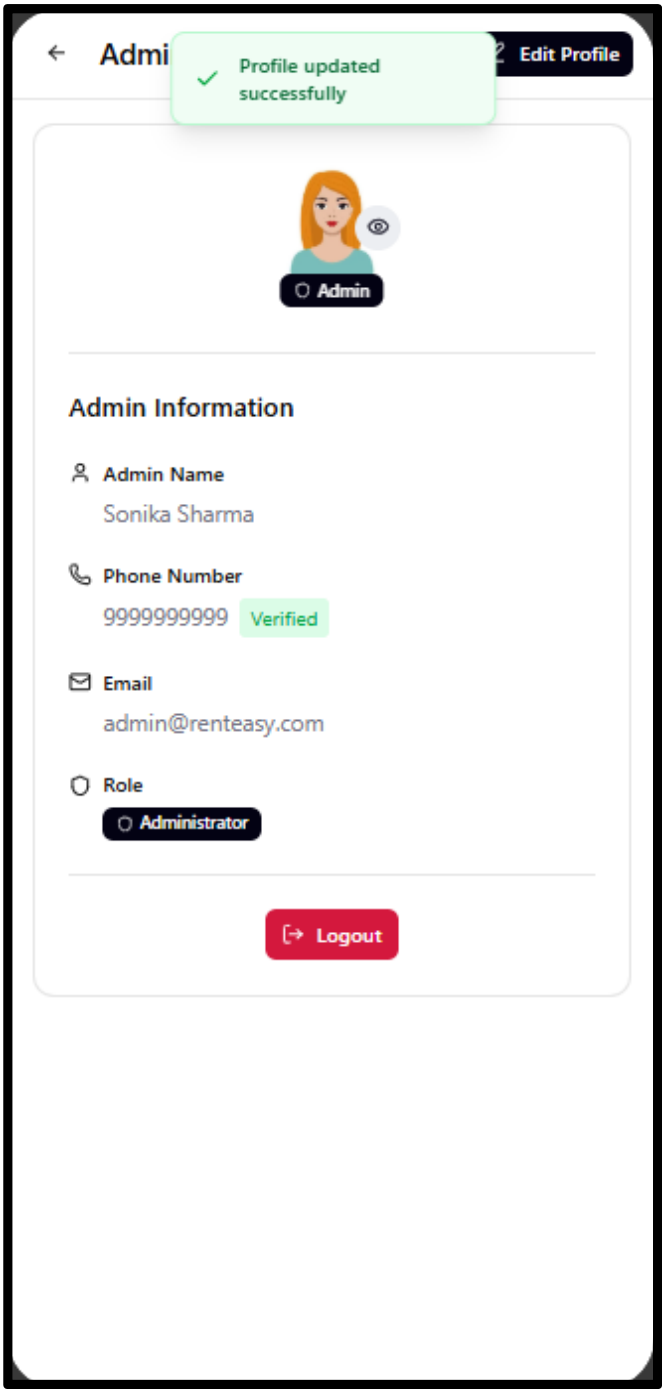
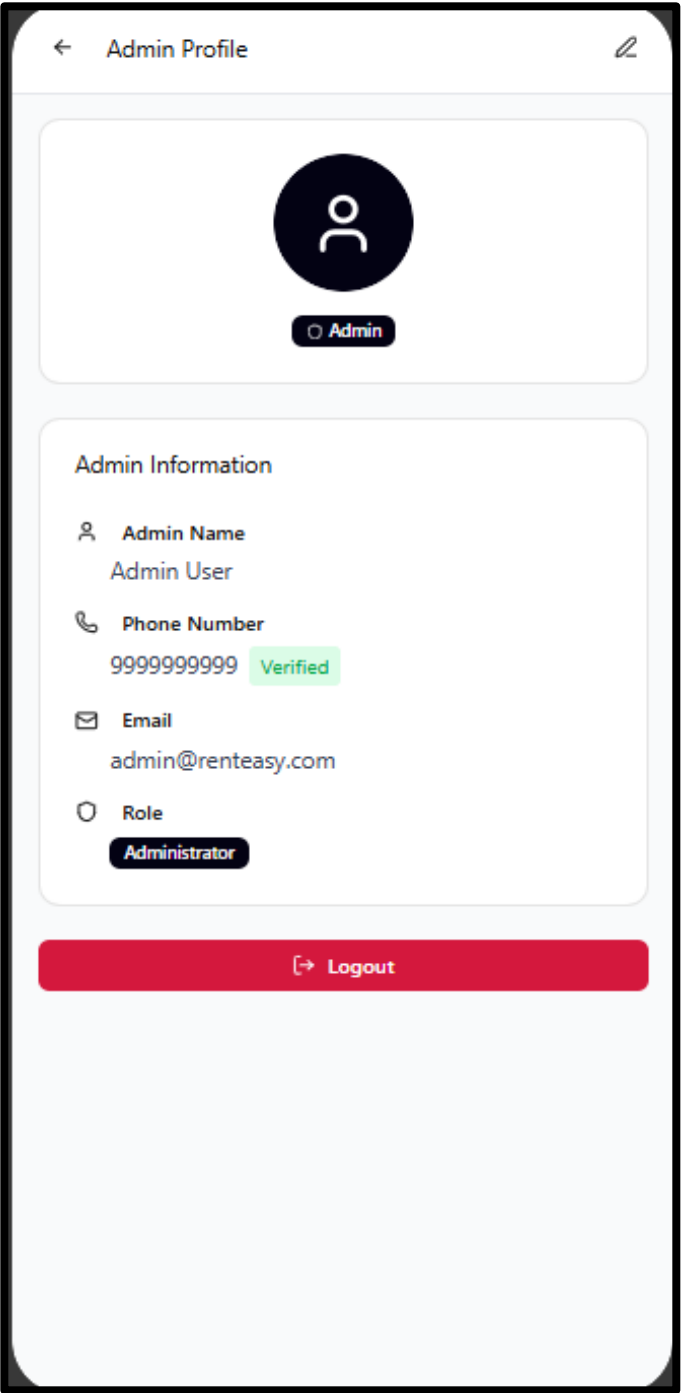
Approve

Reject

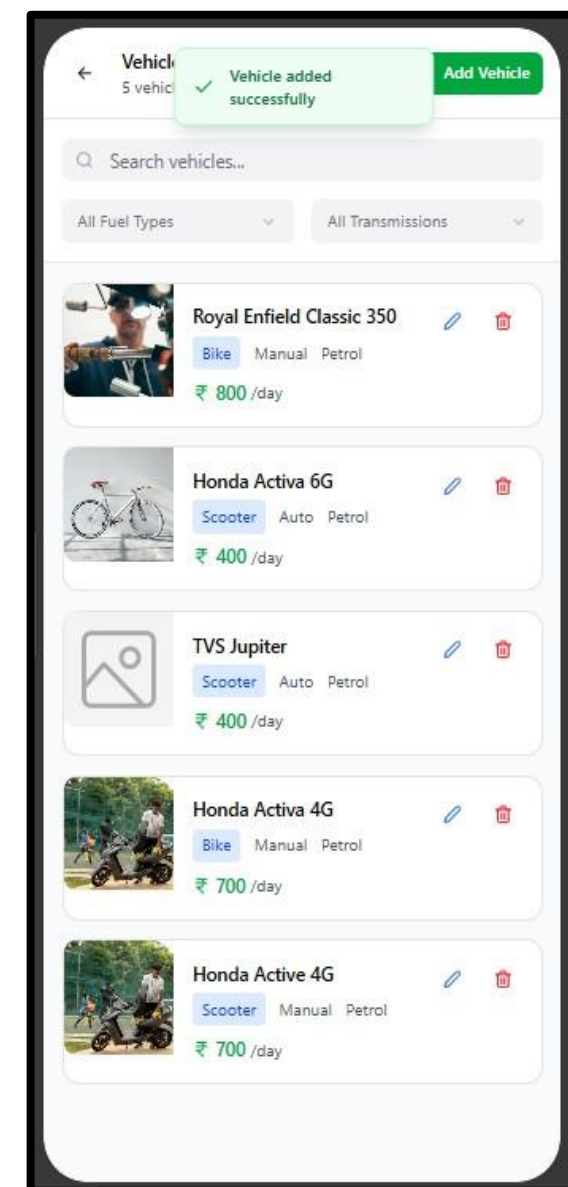
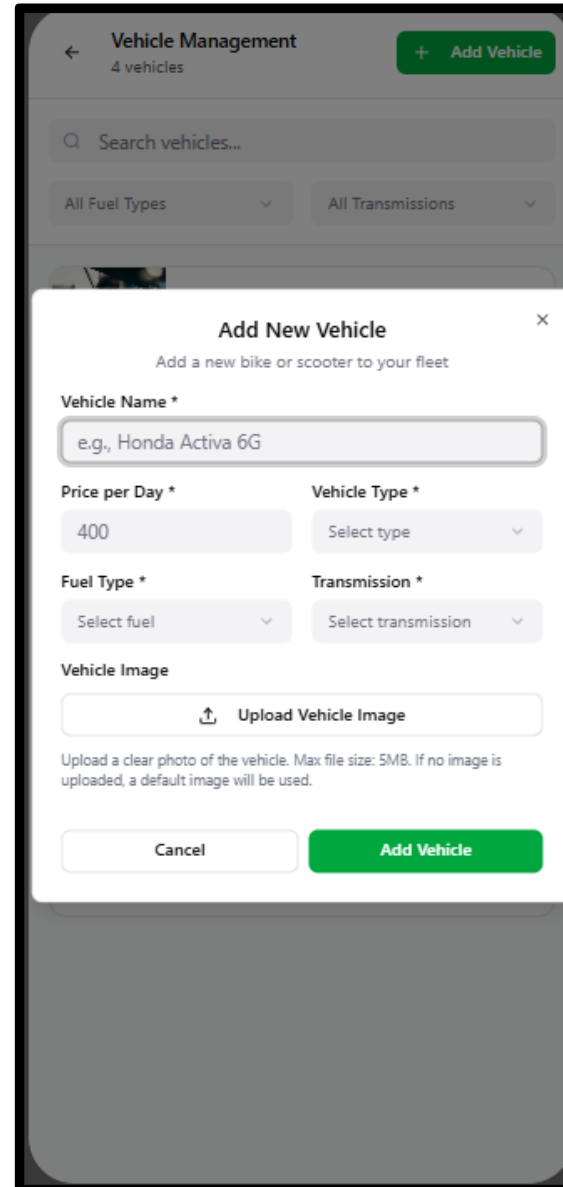
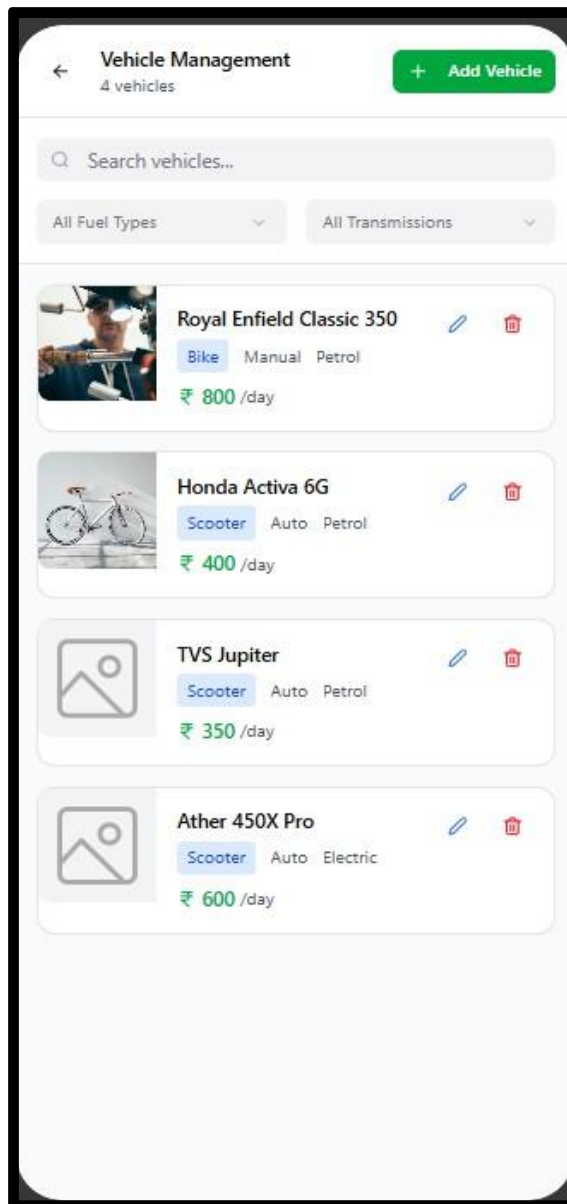


View Bookings

ADMIN PROFILE MANAGEMENT



ADD NEW VEHICLE



EDIT AND DELETE VEHICLES

Vehicle Management

4 vehicles

Add Vehicle

Search vehicles...

All Fuel TypesAll Transmissions

Edit Vehicle

Update vehicle information

Vehicle Name *

Honda Activa 6G

Price per Day *

400

Vehicle Type *

Scooter

Fuel Type *

Petrol

Transmission *

Auto

Vehicle Image



Upload New Image

Upload a clear photo of the vehicle. Max file size: 5MB

Cancel

Save Changes

Vehicle Management

5 vehicles

Vehicle updated successfully

Add Vehicle

Search vehicles...

All Fuel TypesAll Transmissions



Royal Enfield Classic 350

Bike Manual Petrol

₹ 800 /day



Honda Activa 6G

Scooter Auto Petrol

₹ 400 /day



TVS Jupiter

Scooter Auto Petrol

₹ 400 /day



Ather 450X Pro

Scooter Auto Electric

₹ 600 /day



Honda Activa 4G

Bike Manual Petrol

₹ 700 /day

Vehicle Management

5 vehicles

Add Vehicle

Search vehicles...

All Fuel TypesAll Transmissions



Royal Enfield Classic 350

Bike Manual Petrol

₹ 800 /day

Delete Vehicle

Are you sure you want to delete "Honda City"? This action cannot be undone.

Delete

Cancel



Ather 450X Pro

Scooter Auto Electric

₹ 600 /day



Honda Activa 4G

Bike Manual Petrol

₹ 700 /day

Vehicle Management

4 vehicles

Vehicle deleted successfully

Add Vehicle

Search vehicles...

All Fuel TypesAll Transmissions



Royal Enfield Classic 350

Bike Manual Petrol

₹ 800 /day



Honda Activa 6G

Scooter Auto Petrol

₹ 400 /day



TVS Jupiter

Scooter Auto Petrol

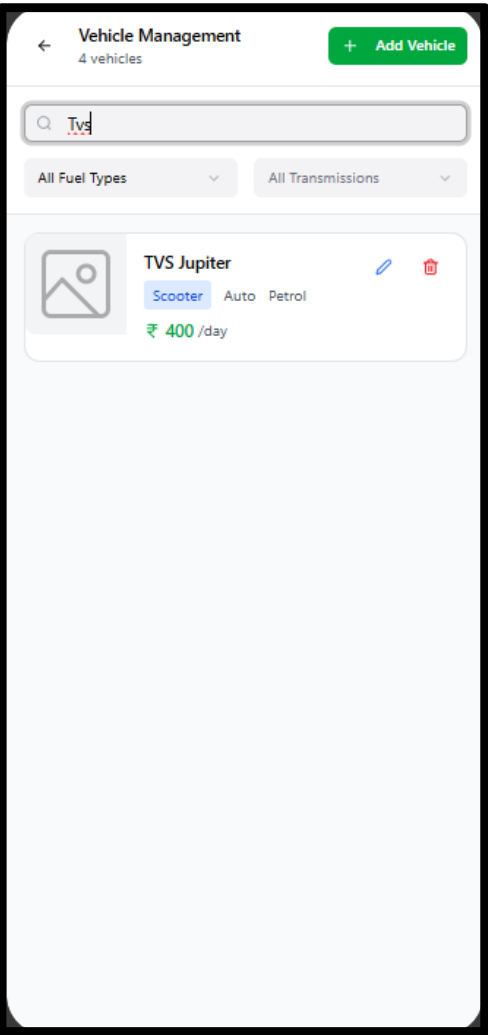
₹ 400 /day



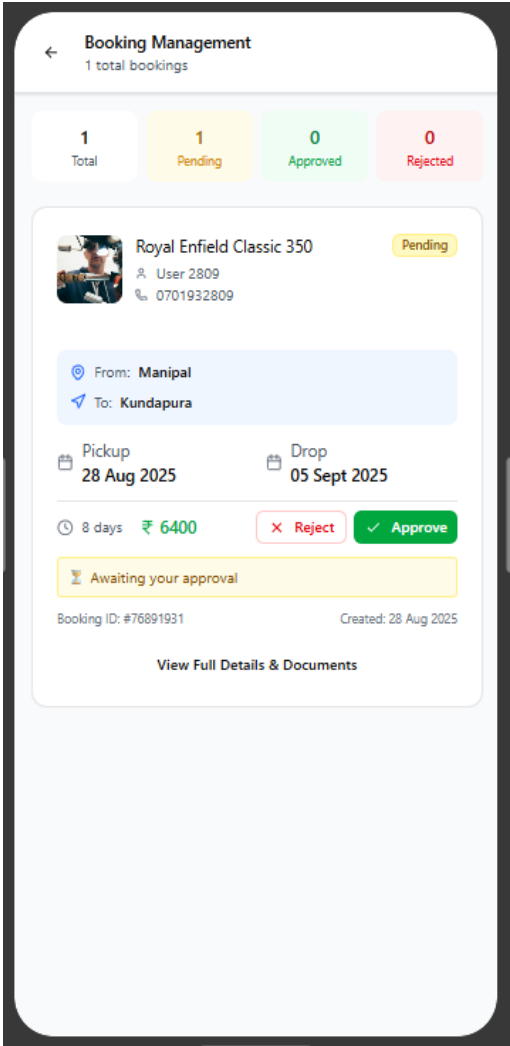
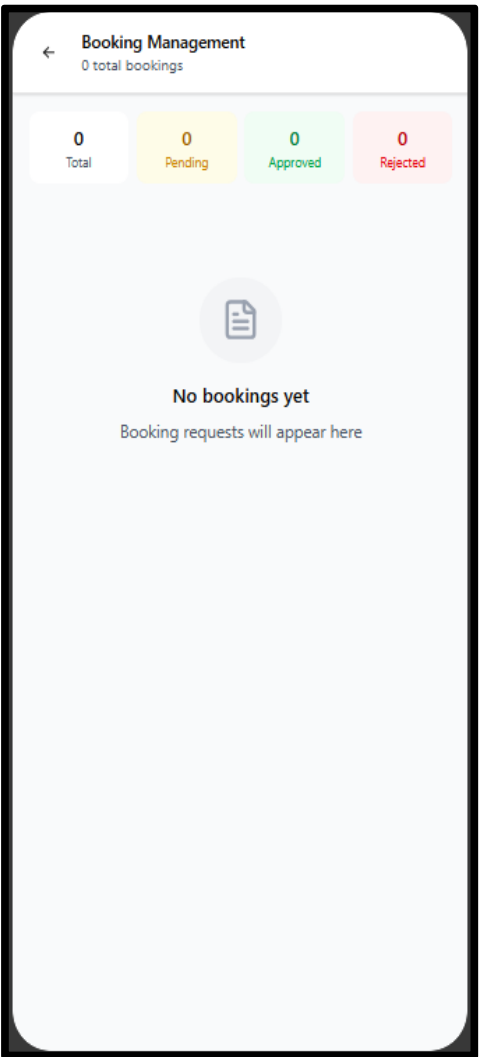
Honda Activa 4G

Bike Manual Petrol

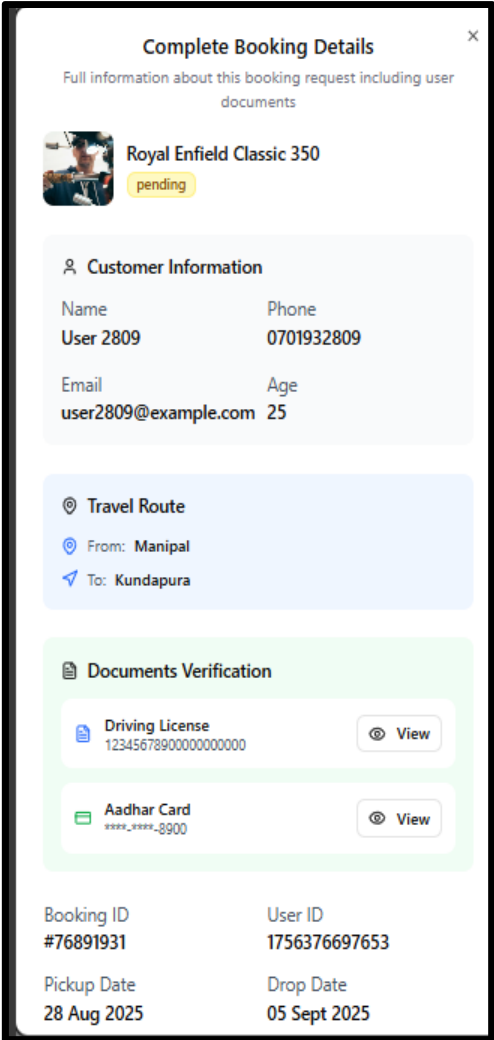
₹ 700 /day



SEARCH



BOOKING MANAGEMENT



Customer Information

Name

User 2809

Phone

0701932809

Email

user2809@example.com

Age

25

Travel Route

From: Manipal

To: Kundapura

Documents Verification

Driving License

12345678900000000000

View

Aadhar Card

****-****-8900

View

Booking ID

#76891931

User ID

1756376697653

Pickup Date

28 Aug 2025

Drop Date

05 Sept 2025

Duration

8 days

Total Amount

₹6400

Created On

28 Aug 2025

Reject

Approve

Customer Information

Name

User 2809

Phone

0701932809

Email

user2809@example.com

Age

25

Travel Route

From: Manipal

To: Kundapura

Approve Booking

Are you sure you want to approve this booking request? The customer will be notified and can proceed with their rental.

Cancel

Approve

Booking ID

#76891931

User ID

1756376697653

Pickup Date

28 Aug 2025

Drop Date

05 Sept 2025

Duration

8 days

Total Amount

₹6400

Created On

28 Aug 2025

Reject

Approve

Bookin

1 total b

Booking approved successfully

1

Total

0

Pending

1

Approved

0

Rejected

Royal Enfield Classic 350

Approved

User 2809

0701932809

From: Manipal

To: Kundapura

Pickup

28 Aug 2025

Drop

05 Sept 2025

8 days

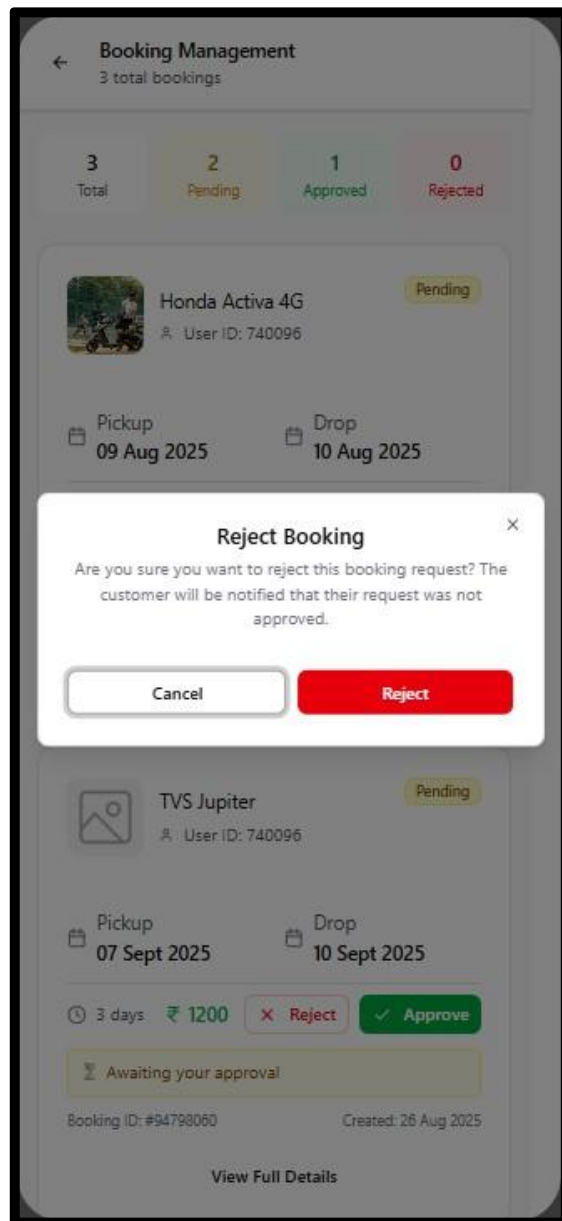
₹ 6400

Booking ID: #76891931

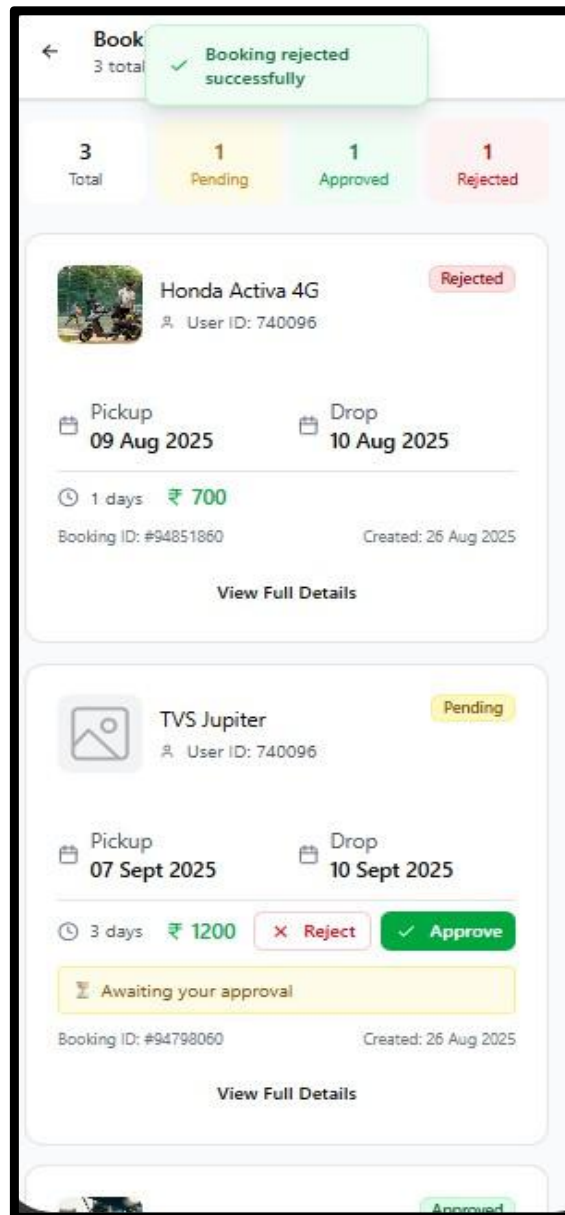
Created: 28 Aug 2025

View Full Details & Documents

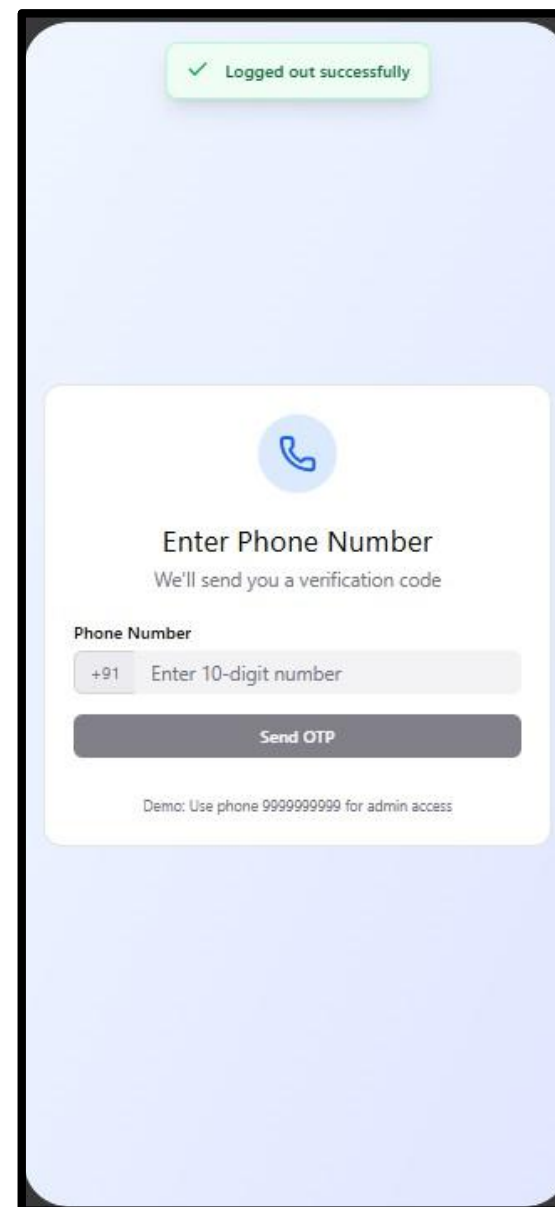
APPROVE BOOKING



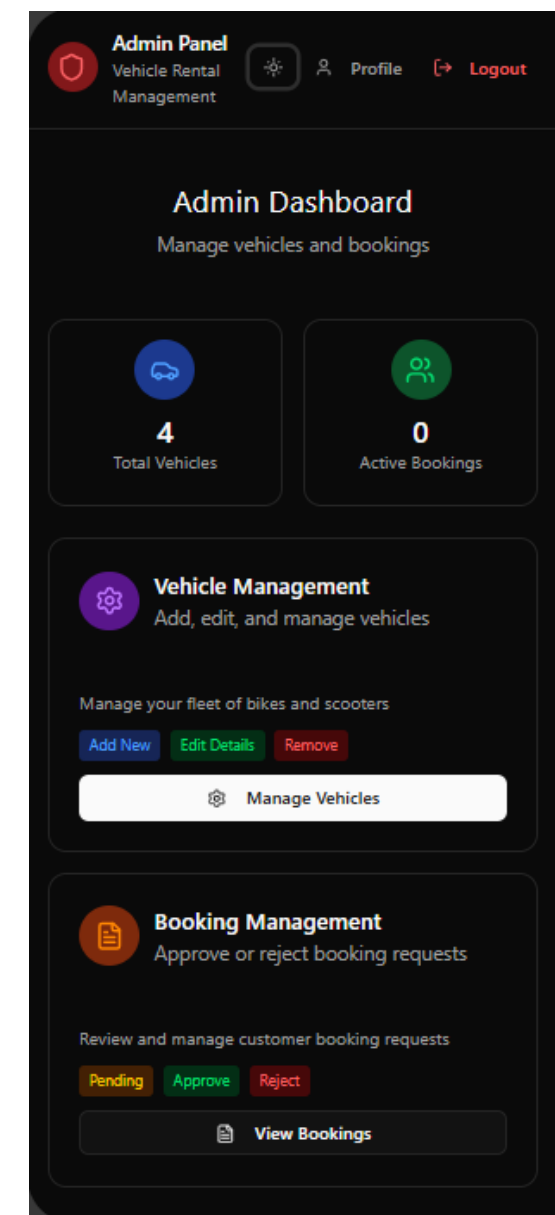
REJECT BOOKING



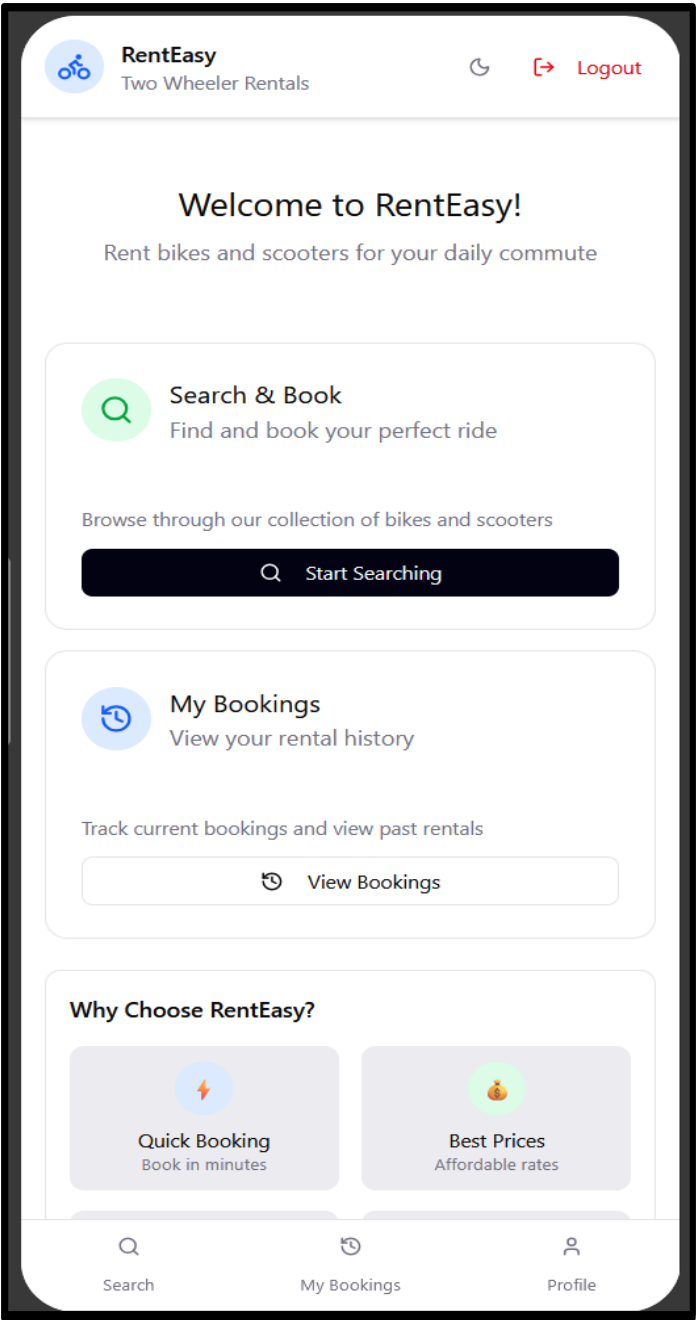
LOGOUT



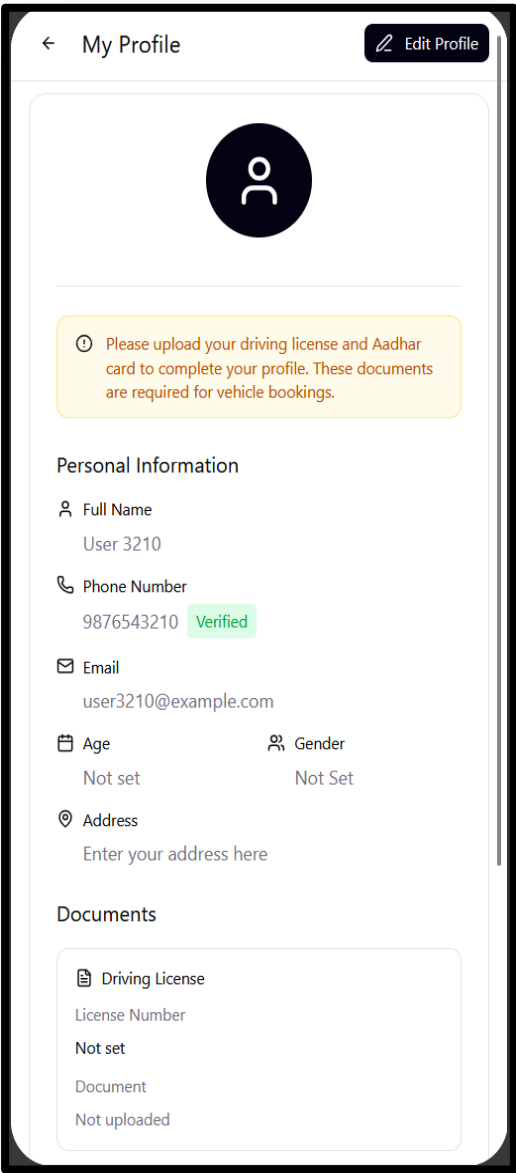
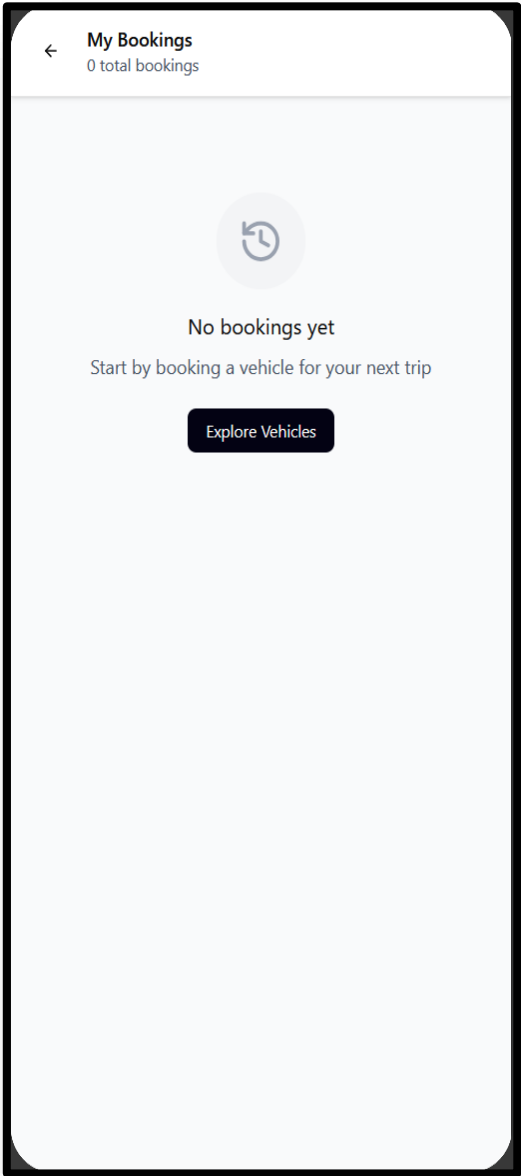
DARKMODE



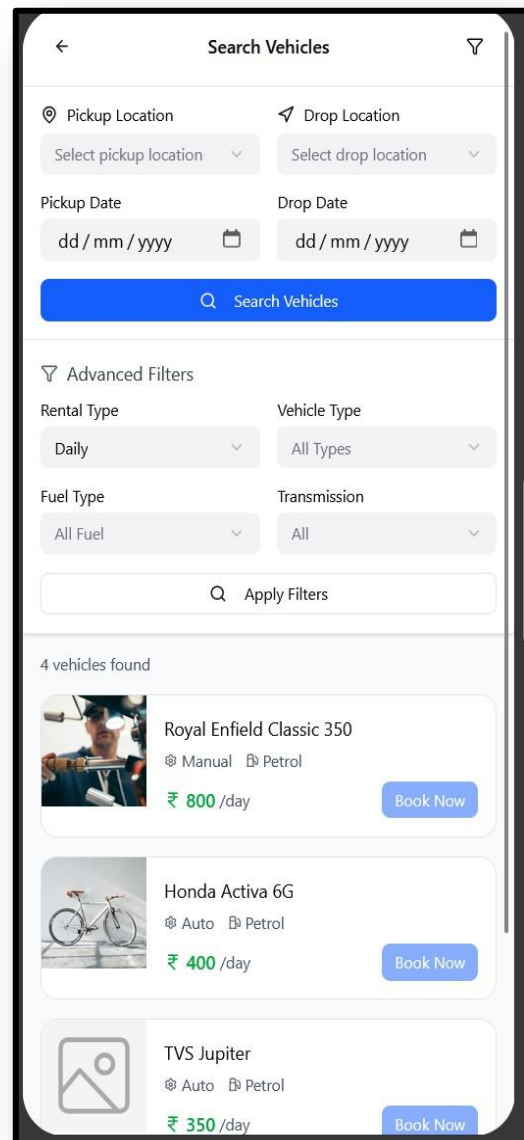
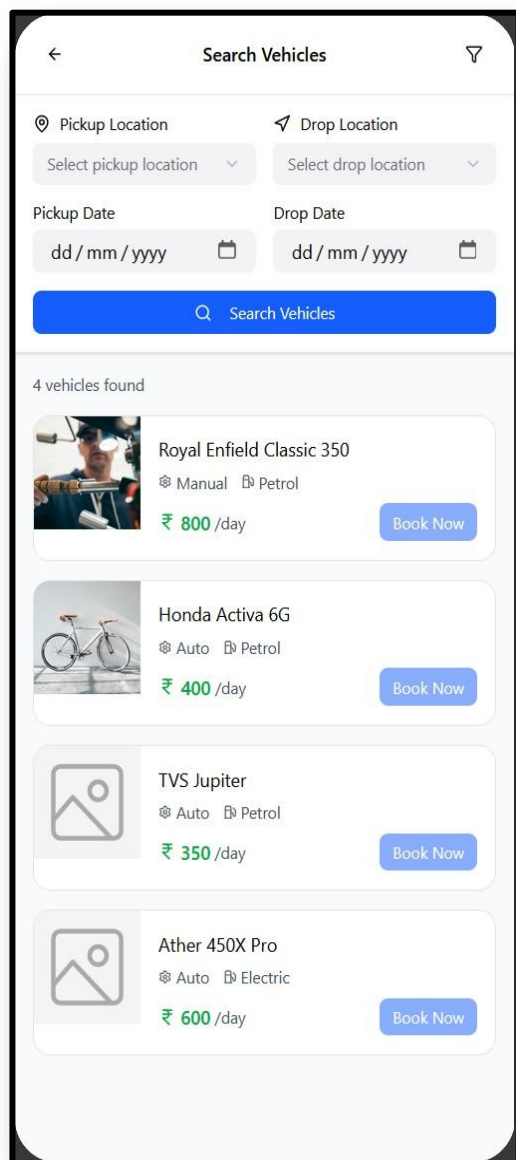
USER HOME PAGE



USER BOOKINGS AND PROFILE



SEARCH VEHICLES AND FILTER



Search Vehicles

Pickup Location

Drop Location

Manipal

Kundapura

Pickup Date

Drop Date

30-08-2025

04-09-2025

🔍 Search Vehicles

4 vehicles found

Manipal → Kundapura

Royal Enfield Classic 350

Manual

Petrol

₹ 800 /day

Book Now

Honda Activa 6G

Auto

Petrol

₹ 400 /day

Book Now

TVS Jupiter

Auto

Petrol

₹ 350 /day

Book Now

Ather 450X Pro

Auto

Electric

₹ 600 /day

Book Now

PAYMENT MODULE

← Complete Payment

Document Verification

ⓘ Please upload your driving license and Aadhar card in your profile before proceeding with payment.

Go to Profile

Booking Summary



Royal Enfield Classic 350

Manual • Petrol

From: Manipal

To: Kundapura

Pickup Date: 28 Aug 2025

Drop Date: 01 Sept 2025

Duration: 4 days

Rate per day: ₹800

Number of days: 4

Total Amount: ₹ 3200

✓ Pay ₹3200

This is a demo payment. No real transaction will be processed.

← My Profile

✕ Cancel

Save

Email

abc@example.com

Age

21

Gender

Female

Address

Enter your address here

Documents

Driving License

License Number

ASD455576578

Document

Upload

Upload a clear photo of your driving license. Max file size: 5MB

Aadhar Card

Aadhar Number

Enter 12-digit Aadhar number

Document

Upload

Upload a clear photo of your Aadhar card. Max file size: 5MB

Logout

← My Profile

✕ Cancel

Save

Age

21

Gender

Female

Address

Enter your address here

Documents

Driving License

License Number

ASD455576578

Document

View

Upload

Upload a clear photo of your driving license. Max file size: 5MB

Aadhar Card

Aadhar Number

1234357875

Document

View

Upload

Upload a clear photo of your Aadhar card. Max file size: 5MB

Documents Complete

Your profile is ready for vehicle bookings

Logout

Z

CONFIRM PAYMENT

<

Select Payment Method

Payment Method

Credit/Debit Card

UPI

Net Banking

Order Summary

Royal Enfield Classic 350

Duration: 4 days

Total Amount

₹3200

₹3200

Pay ₹3200

<

Payment Successful

Payment Successful!

Your booking has been confirmed

Transaction Details

Transaction ID: TXN1IQG7GXVT

Amount Paid: ₹3200

Payment Method: Credit Card

Date & Time: 28 Aug 2025, 07:41 pm

Booking Details

Royal Enfield Classic 350

Manual • Petrol

From: Manipal

To: Kundapura

Pickup: 28 Aug 2025

Drop: 01 Sept 2025

Duration: 4 days

Important Information

• Please carry your driving license and uploaded documents

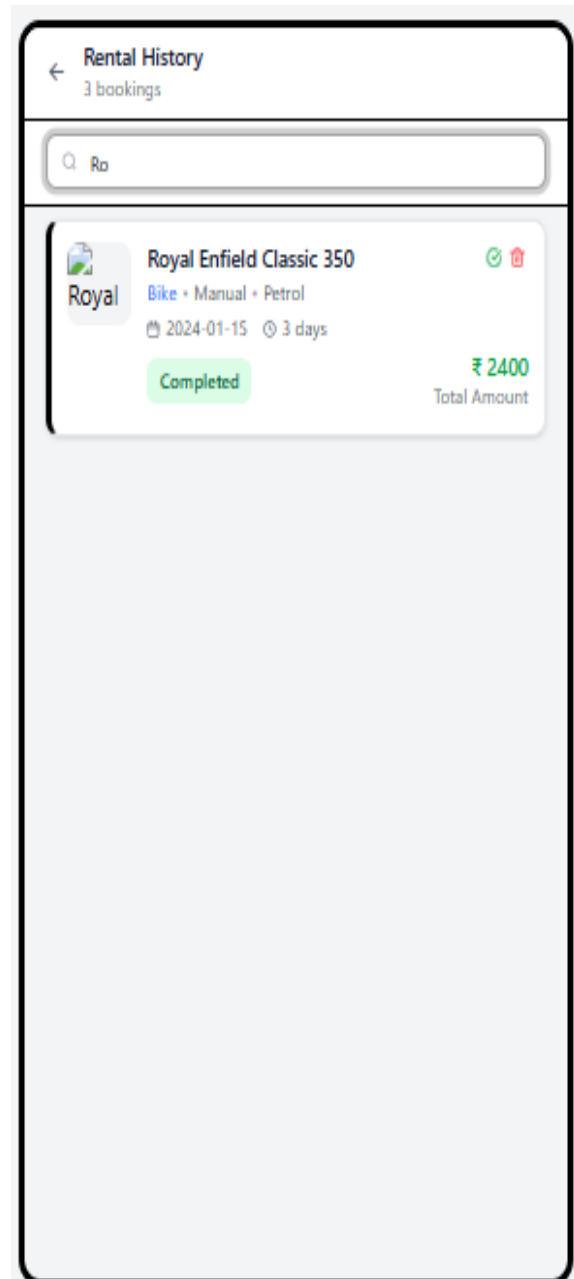
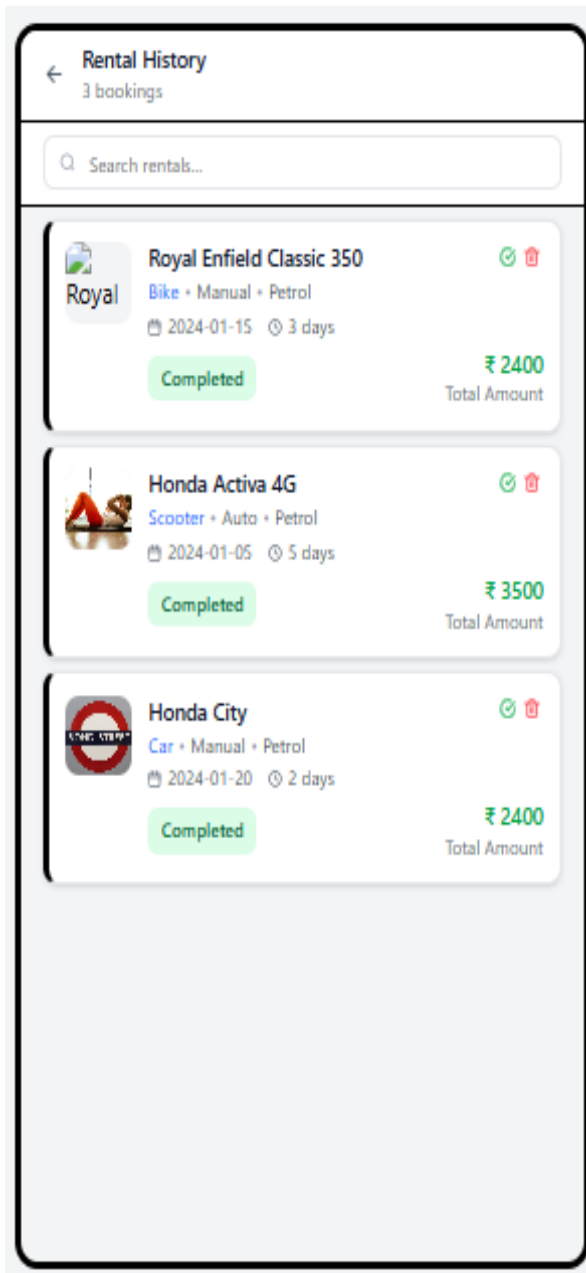
• Arrive 15 minutes before pickup time

• Support: +91-9876543210 | help@bikereental.com

Download Receipt

Book Another Ride

USER HISTORY MODULE



PARTICULAR HISTORY DETAILS

← Rental History

3 bookings

🔍 Search rentals...



Royal Enfield Classic 350

Bike • Manual • Petrol

📅 2024-01-15 ⌚ 3 days

Completed

₹ 2400

Total Amount



Honda Activa 4G

Scooter • Auto • Petrol

📅 2024-01-05 ⌚ 5 days

Completed

₹ 3500

Total Amount



Honda City

Car • Manual • Petrol

📅 2024-01-20 ⌚ 2 days

Completed

₹ 2400

Total Amount

← Rental Details

Completed



Honda City

Car • Manual • Petrol

₹ 2400

Total Amount

Booking Information

📅 Duration

2024-01-20 to 2024-01-22

⌚ Period

2 days

Customer Information

👤 Priya Sharma

📞 +91 87654 32109

✉️ priya.sharma@email.com

Trip Completed

Trip Route

📍 From: Kundapur

📍 To: Manipal

📄 View Bill

← Bill Details

RENTAL INVOICE

Invoice #INV-4

Date: 2024-01-20



Honda City

Car • Manual • Petrol

Customer Details

Rental Period

👤 Priya Sharma

+91 87654 32109

📅 Pickup: 2024-01-20

📅 Drop: 2024-01-22

⌚ 2 day(s)

Location Details

📍 From: Kundapur

📍 To: Manipal

Bill Breakdown

Rental Cost (2 days × ₹1200)

₹ 2400

Total Paid

₹ 2400

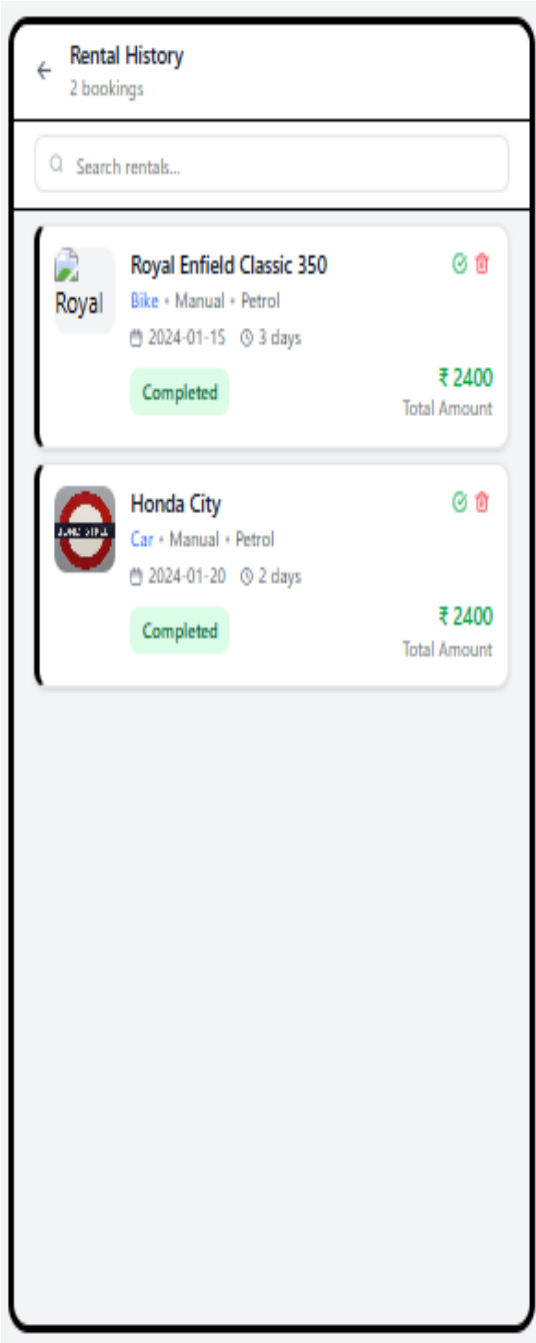
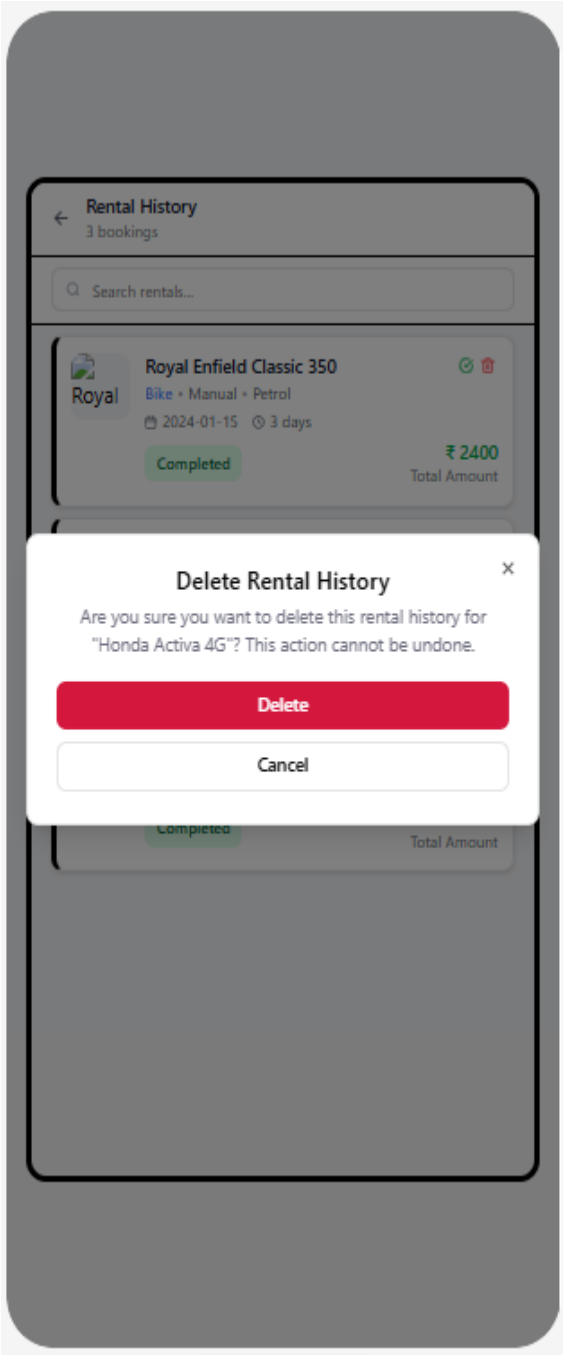
Payment Information

💳 Paid via UPI - Success

Transaction ID: TXN4123456

Download PDF Invoice

DELETING PARTICULAR HISTORY



LOGOUT

✓ Logged out successfully



Enter Phone Number

We'll send you a verification code

Phone Number

+91 Enter 10-digit number

Send OTP

Demo: Use phone 9999999999 for admin access

Abstract:

The *Vehicle Rental Management System* is a mobile-based application designed to streamline the process of vehicle rentals through a secure, efficient, and user-friendly interface. Leveraging Firebase Authentication, Realtime Database, and Storage, the system offers real-time data handling, cloud-based storage, and OTP-based login for robust security. It caters to both users and administrators, enabling seamless vehicle search, booking, inventory management, and real-time status tracking. By minimizing manual intervention and enhancing the user experience, this system provides a modern solution for urban vehicle rental needs.

1. Introduction

The Vehicle Rental Management System is a mobile-based platform that simplifies the process of renting vehicles for customers and managing vehicle inventory for administrators. It provides a secure OTP-based authentication system, a user-friendly interface for vehicle search and booking, and an admin dashboard for monitoring and approving rental requests. The system integrates Firebase Authentication for secure logins, Firebase Realtime Database for storing booking and vehicle data, and Firebase Storage for hosting vehicle images.

For users, the application offers features to search for vehicles based on location, rental type, and availability, as well as track the status of their bookings. For administrators, it provides tools to add, edit, or remove vehicles, review booking requests, and update their status in real time. This dual-role architecture ensures a smooth, efficient, and secure rental experience for all parties.

2. Purpose

The primary purpose of this project is to develop a secure and efficient platform that facilitates easy access to vehicle rental services via a mobile application. The system aims to:

- Provide a seamless user experience for discovering and booking vehicles.
- Equip administrators with tools for managing vehicle listings and responding to rental requests in real time.

- Ensure data integrity and user security using Firebase services.
- Reduce manual processes and improve rental service transparency and speed.
- Demonstrate the viability of cloud-integrated, mobile-first solutions for rental service industries.

3. Core Objectives

- Secure Authentication: Implement OTP-based login via Firebase Authentication to ensure safe and quick sign-in for both users and admins.
- Role-Based Access Control: Automatically redirect users or admins to their respective dashboards based on their role in the Firebase Database.
- Vehicle Discovery: Allow users to search vehicles by rental type, city, pickup date/time, and drop date/time.
- Booking Workflow: Enable users to view details of available vehicles, initiate bookings, and simulate payments.
- Admin Vehicle Management: Allow admins to manage vehicle listings with the ability to add, edit, or delete entries.
- Booking Management: Provide admins with real-time access to booking requests for approval or rejection.
- Firebase Storage Integration: Store vehicle images in the cloud and link them to database entries for dynamic display.
- Real-Time Updates: Notify users instantly when booking status changes.
- Session Management: Maintain persistent sessions for logged-in users and admins, with a secure logout option.

4. Detailed Module Breakdown

- Module 1: User Authentication Module

Main Function: Securely authenticate users and admins via phone number and OTP.

Sub-Functionalities:

- OTP sent and verified using Firebase Authentication.
- Role verification from Firebase Realtime Database (role = user or admin).
- Session persistence until logout.
- Redirection to user or admin dashboard based on role.

User Flow: The user enters their phone number, receives an OTP, and verifies it. The app checks the user's role in the database and redirects to the correct dashboard.

- Module 2: Vehicle Search & Booking (User Side)

Main Function: Allow users to find and book vehicles according to their needs.

Sub-Functionalities:

- Search filters for Rental Type (Daily/Monthly), City, Pickup Date/Time, and Drop Date/Time.
- Display of matching vehicles in scrollable cards showing name, transmission, fuel type, rent/day, and image.
- "Book Now" button to initiate booking process.
- Store booking request in Firebase with status = pending.
- Redirect user to dummy payment screen.

User Flow: After logging in, the user selects desired filters, views the results, chooses a vehicle, and proceeds with booking.

- Module 3: Booking Details (User Side)

Main Function: Provide users with a clear overview of all their bookings.

Sub-Functionalities:

- List of bookings with details: Vehicle Name, Booking Dates, Duration, Total Rent.
- Real-time status updates: Pending / Approved / Rejected.
- Notifications on status change by admin.

User Flow: The user taps "My Bookings" from the dashboard to view and track all bookings.

- Module 4: Dummy Payment Flow (User Side)

Main Function: Simulate payment processing after booking.

Sub-Functionalities:

- Direct user to a simulated payment page.
- On confirmation, payment is marked as simulated, and booking remains pending until approved.
- Redirect user to "My Bookings" after simulated payment confirmation.

- Module 5: Admin Authentication Module

Main Function: Grant secure admin access with the same OTP-based login as users.

Sub-Functionalities:

- OTP verification using Firebase Authentication.
- Role validation from Firebase Database.
- Redirection to Admin Dashboard for authorized admins only.

User Flow: Admin enters phone number, verifies OTP, and is redirected to admin dashboard upon successful role validation.

- Module 6: Vehicle Management (Admin Side)

Main Function: Enable admins to manage vehicle listings.

Sub-Functionalities:

- Add new vehicles with details: Name, Rent Price, Fuel Type, Transmission, and Images.
- Upload vehicle images to Firebase Storage and store image URLs in the database.
- Edit vehicle details directly from the dashboard.
- Delete outdated or unavailable vehicles.

User Flow: Admin navigates to “Vehicle Management” and chooses to add, edit, or remove vehicles.

- Module 7: Booking Management (Admin Side)

Main Function: Allow admins to handle user booking requests.

Sub-Functionalities:

- View all pending bookings with user ID, vehicle, duration, amount, and status.
- Approve or reject bookings by updating status in Firebase (approved / rejected).
- Automatic real-time updates to user’s “My Bookings” page.

User Flow: Admin reviews each booking and updates its status.

- Module 8: Firebase Storage Integration

Main Function: Manage cloud storage for vehicle images.

Sub-Functionalities:

- Store images uploaded by admin in Firebase Storage.
- Save the public URL of each image in the database.
- Retrieve and display images dynamically in search results and booking pages.

- Module 9: Logout Module

Main Function: Securely log out any user or admin.

Sub-Functionalities:

- Clear active session.
- Redirect to login page.
- Prevent access to dashboard after logout.

5. Technologies Used to implement

Development Platform: Android Studio

Programming Language: Java (Android)

Backend Services: Firebase Authentication (OTP login), Firebase Realtime Database (Vehicle & booking data), Firebase Storage (Vehicle images)

UI Design: Android Material Design Components, RecyclerView for vehicle listings & booking history

Session Management: Firebase Auth sessions + SharedPreferences for basic settings

Push Notifications: Firebase Cloud Messaging (Booking status updates)

6. Conclusion

The Vehicle Rental Management System provides an all-in-one solution for renting vehicles online with minimal manual intervention. Its OTP-based login ensures security, while Firebase integration offers real-time data management. The platform benefits users by providing quick and easy vehicle search and booking, and it benefits admins by giving them efficient tools to manage inventory and approve/reject bookings instantly. This project demonstrates the potential of mobile-based rental platforms in delivering convenience, security, and efficiency to both service providers and customers.